UGANDA ADVANCED LEVEL CERTIFICATE

END OF TERM ONE ASSESSMENT

Chemistry Paper 1

(Theory)

Duration: 2hours 45 minutes

Senior Five

Instructions to candidates:

The paper consists of two sections A and B.

Sections A: *consists of TWO compulsory items***.**

Section B: *consists of two parts I and II each having two items. Attempt any* ***ONE*** *item from each part of section B.*

Responses to these items are to be written in the spaces provided in this booklet. Use ***blue*** *or* ***black*** *ink. All working* ***must*** *be clearly shown.*

Graph paper is provided.

Mathematical tables and silent non-programmable scientific calculators may be

Where necessary use the following: Molar gas constant, R = 8.31JK^{-1}mol^{-1} Molar volume of a gas at s.t.p. = 22.4litres Standard temperature = 273K Standard pressure = 10125Nm^{-1}

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Section A

(Attempt all items)

ITEM 1:

A radiopharmaceutical production facility has received two zirconium-containing samples intended for the preparation of **Zirconium-89**. This isotope is required for labelling monoclonal antibodies used in **Positron Emission Tomography**, where accurate tumour localisation depends on a sufficiently long-lived positron-emitting tracer.

Both samples are considered for use in the synthesis of radiolabelled antibodies, but the production team must evaluate which provides a higher proportion of the required radioisotope, taking into account that the effectiveness of PET imaging depends on the availability of sufficient Zirconium-89, its decay characteristics, and compatibility with the biological half-life of antibodies used in Positron Emission Tomography.

- **Sample A:** RAM = 91.09
- **Sample B:** RAM = 90.60

Mass spectrometry has provided the isotopic composition of the two samples:

Isotope (mass number)	Abundance (%) in both samples
90	Unknown
89	Unknown
92	17.1
94	17.4
96	2.8

The radioactive isotope needed is **Zirconium-89** a key component, but its abundance in each sample is unknown.

Task:

a) Explain how the abundances of the different isotopes in the sample were obtained.

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b) Determine the percentage of **Zr-89** in the two samples and provide a recommendation on which of the two can be used in the labeling of antibodies.

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c) Suggest two other ways in which the relative atomic mass of elements can be determined.

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ITEM 2:

In a nuclear medicine laboratory in Kampala, a team of medical physicists is evaluating a newly prepared sample of the synthetic radioisotope technetium-99 (^{99}Tc), which is widely used in diagnostic imaging due to its beta emission properties and suitable decay characteristics.

To assess the decay behaviour of the isotope, the team measures the activity of the sample at different time intervals after preparation. The recorded disintegration rates (in disintegrations per minute) at various times are shown below:

Support material:

Time (hours)	Disintegrations per minute (dpm)
2.5	130
5.0	104
7.5	77
10.0	59
12.5	46
17.5	24

The team is interested in understanding how the activity of the isotope changes over time and whether it follows an exponential decay pattern. They are instructed to analyze the data by transforming the disintegration rates using a logarithmic scale and then plotting the natural logarithm of the activity against time.

Task:

As a chemistry learner, you have been asked to assist;

a) Analyze the data by plotting a logarithmic graph of mass against time. Use the resulting relationship to determine the isotope's half-life and its initial mass, as this is important for understanding the isotope's viable time window for medical imaging applications.

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b) Given that in the nuclear reactor, the isotope was found to undergo three successive beta-emission decays, Write the nuclear decay equation to help them identify the nuclide formed.

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c) Suggest at least three applications of radioactive decay.

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SECTION B:
PART I:
(Attempt only one item from this part)

ITEM 3:

A pain-relief medication has recently been widely circulated in the community. Following reports of misuse and addiction, several patients have been admitted to hospital with symptoms of withdrawal and relapse complications**.** One patient, Betty**,** was admitted after a relapse episode. During clinical questioning, she denied consuming the suspected addictive drug and instead claimed that she had only taken a common analgesic, **acetaminophen (paracetamol)**.

A 2.50 g sample of the unknown substance was completely combusted in excess oxygen, producing 5.83 g of carbon dioxide and 1.34g of water produced. In another combustion experiment 0.80 g was done at r.t.p producing 63.0 cm^3 of nitrogen.

The vaporized sample was found to have a relative vapour density 75.8.

Support material:

Painkiller	**Molecular formula**
Acetaminophen(paracetamol)	$C_8H_9NO_2$
Oxycodone	$C_{18}H_{21}NO_4$

Task:

As a chemistry learner, determine the molecular formula of the drug from the given analytical data, and evaluate whether the patient's relapse was caused by the resumption of the addictive painkillers. In your response, also explain how

the drug functions in the body and discuss the potential effects and impacts associated with its use.

ITEM 4:

An industrial company has purchased gas cylinders from a supplier for use in industrial heating processes. The cylinders are labelled as containing a hydrocarbon fuel with more than four carbon atoms, expected to produce high heat energy on combustion.

However, during initial use, one worker reports that the gas cylinder feels unusually light and the flame produced appears weak and insufficient for heating purposes. This raises suspicion that the gas may not be the expected hydrocarbon (such as butane or higher alkanes). To confirm the identity and purity of the gas, the company submits a sample to the laboratory for eudiometric combustion analysis.

A 30 cm^3 sample of a hydrocarbon gas is mixed with excess oxygen in a eudiometer tube and ignited to ensure complete combustion.

After combustion, the mixture is allowed to cool back to room temperature. The remaining gases are then passed through a solution of sodium hydroxide (NaOH), and a reduction in gas volume is observed. The decrease in volume after treatment with NaOH is 90 cm^3.

The vapour density of the hydrocarbon was determined to be 1.88 g/l at standard temperature and pressure (STP).

Task:

As a chemistry learner, determine the molecular formula of the hydrocarbon and identify the homologous series to which it belongs. In addition, explain the possible environmental impacts associated with the use of carbon-based fuels.

PART II

(Attempt only one item from this part)

ITEM 5:

A biomedical startup operating in Nairobi set out to produce portable glucose biosensors for use in regional clinics. The design relied on a transducer layer

made of graphene due to its high electron affinity and excellent electrical conductivity.

In the device, glucose oxidase was immobilized on the graphene surface. During operation, glucose in a blood sample was oxidized, releasing electrons. The graphene layer efficiently accepted these electrons and transferred them to the electrode, producing a current directly proportional to glucose concentration. Challenges emerged during scale-up. The production team encountered difficulties sourcing high-quality graphene and carbon nanotubes in sufficient quantities.

However, Racheal, a Senior Five student, proposed using Period 2 elements on the surface of the biosensor. After obtaining electron affinity data, she was unable to properly interpret the information and sought your assistance to help analyze it and improve her presentation.

Support material:

Element	**Li**	**Be**	**B**	**C**	**N**	**O**	**F**	**Ne**
Electron affinity (kJ/mol	**-60**	**+18**	**-27**	**-122**	**+7**	**-141**	**-328**	**+29**

Task:

As a learner of chemistry, Analyse the data through plotting a linear graph of electron affinity against the elements and explain the trend observed. Identify the most effective element for surface use and explain your choice. Conclude by discussing the relationship between electron affinity and the reactivity of non-metallic elements.

ITEM 6:

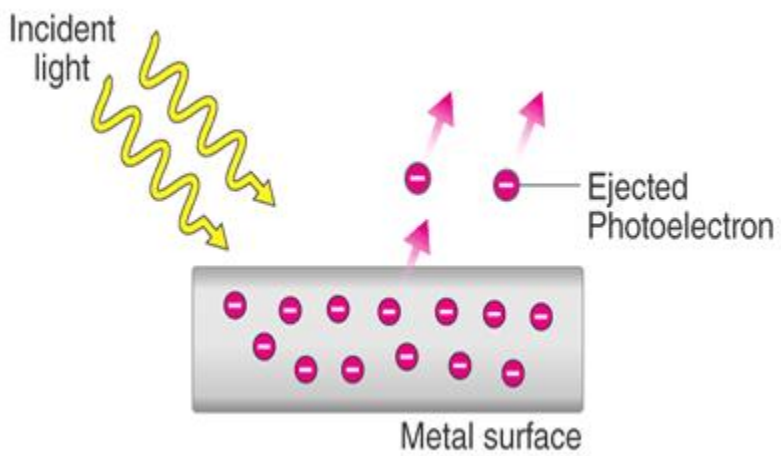
Motion and position sensors used in modern automation systems rely on the photoelectric effect. In these systems, a **light-emitting diode (LED)** (UV or IR source) is directed toward a photoelectric material **connected** to an electronic circuit.

When an object passes between the LED and the sensor, the light beam is interrupted. This causes a change in electron emission from the photoelectric surface, resulting in a detectable change in **potential difference**, which is used to register motion or position.

A physics laboratory is investigating suitable materials for use in such a sensor. A beam of UV light with energy **6.50 eV** is directed at material surfaces made from period three elements **A, B, C, D and E**.

The students observe whether electrons are emitted and relate the observations to the elements to be utilized**.**

Support material:



Element	**Ionisation energy/kJmol^{-1}**					
	1st	**2nd**	**3rd**	**4th**	**5th**	**6th**
A	1,000	2,250	3,360	4,560	7,010	8,500
B	1,012	1,900	2,910	4,960	6,270	22,200
C	738	1,450	7730	10,542	13,630	18,020
D	496	4,560	6190	9,543	13,354	16,613
E	786	1,580	3230	4,360	16,100	19,805
F	578	1820	2750	11,600	14,841.9	18,379

Task:

As a chemistry learner, determine the identities of elements A–E by analyzing their electronic configurations to identify their respective groups and blocks. Explain the factors responsible for the observed differences in their ionisation

energies, and discuss how these trends relate to the reactivity of the metallic elements identified.

END

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Periodic Table of the Elements

1 IA 1A	2 IIA 2A	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 **H** Hydrogen 1.008																	2 **He** Helium 4.003
3 **Li** Lithium 6.941	4 **Be** Beryllium 9.012											5 **B** Boron 10.811	6 **C** Carbon 12.011	7 **N** Nitrogen 14.007	8 **O** Oxygen 15.999	9 **F** Fluorine 18.998	10 **Ne** Neon 20.180
11 **Na** Sodium 22.990	12 **Mg** Magnesium 24.305											13 **Al** Aluminum 26.982	14 **Si** Silicon 28.086	15 **P** Phosphorus 30.974	16 **S** Sulfur 32.066	17 **Cl** Chlorine 35.453	18 **Ar** Argon 39.948
19 **K** Potassium 39.098	20 **Ca** Calcium 40.078	21 **Sc** Scandium 44.956	22 **Ti** Titanium 47.867	23 **V** Vanadium 50.942	24 **Cr** Chromium 51.996	25 **Mn** Manganese 54.938	26 **Fe** Iron 55.845	27 **Co** Cobalt 58.933	28 **Ni** Nickel 58.693	29 **Cu** Copper 63.546	30 **Zn** Zinc 65.38	31 **Ga** Gallium 69.723	32 **Ge** Germanium 72.631	33 **As** Arsenic 74.922	34 **Se** Selenium 78.971	35 **Br** Bromine 79.904	36 **Kr** Krypton 83.798
37 **Rb** Rubidium 85.468	38 **Sr** Strontium 87.62	39 **Y** Yttrium 88.906	40 **Zr** Zirconium 91.224	41 **Nb** Niobium 92.906	42 **Mo** Molybdenum 95.95	43 **Tc** Technetium 98.907	44 **Ru** Ruthenium 101.07	45 **Rh** Rhodium 102.906	46 **Pd** Palladium 106.42	47 **Ag** Silver 107.868	48 **Cd** Cadmium 112.414	49 **In** Indium 114.818	50 **Sn** Tin 118.711	51 **Sb** Antimony 121.760	52 **Te** Tellurium 127.6	53 **I** Iodine 126.904	54 **Xe** Xenon 131.294
55 **Cs** Cesium 132.905	56 **Ba** Barium 137.328	57-71	72 **Hf** Hafnium 178.49	73 **Ta** Tantalum 180.948	74 **W** Tungsten 183.84	75 **Re** Rhenium 186.207	76 **Os** Osmium 190.23	77 **Ir** Iridium 192.217	78 **Pt** Platinum 195.085	79 **Au** Gold 196.967	80 **Hg** Mercury 200.592	81 **Tl** Thallium 204.383	82 **Pb** Lead 207.2	83 **Bi** Bismuth 208.980	84 **Po** Polonium [208.982]	85 **At** Astatine 209.987	86 **Rn** Radon 222.018
87 **Fr** Francium 223.020	88 **Ra** Radium 226.025	89-103	104 **Rf** Rutherfordium [261]	105 **Db** Dubnium [262]	106 **Sg** Seaborgium [266]	107 **Bh** Bohrium [264]	108 **Hs** Hassium [269]	109 **Mt** Meitnerium [278]	110 **Ds** Darmstadtium [281]	111 **Rg** Roentgenium [280]	112 **Cn** Copernicium [285]	113 **Nh** Nihonium [286]	114 **Fl** Flerovium [289]	115 **Mc** Moscovium [289]	116 **Lv** Livermorium [293]	117 **Ts** Tennessine [294]	118 **Og** Oganesson [294]

Lanthanide Series	57 **La** Lanthanum 138.905	58 **Ce** Cerium 140.116	59 **Pr** Praseodymium 140.908	60 **Nd** Neodymium 144.243	61 **Pm** Promethium 144.913	62 **Sm** Samarium 150.36	63 **Eu** Europium 151.964	64 **Gd** Gadolinium 157.25	65 **Tb** Terbium 158.925	66 **Dy** Dysprosium 162.500	67 **Ho** Holmium 164.930	68 **Er** Erbium 167.259	69 **Tm** Thulium 168.934	70 **Yb** Ytterbium 173.055	71 **Lu** Lutetium 174.967
Actinide Series	89 **Ac** Actinium 227.028	90 **Th** Thorium 232.038	91 **Pa** Protactinium 231.036	92 **U** Uranium 238.029	93 **Np** Neptunium 237.048	94 **Pu** Plutonium 244.064	95 **Am** Americium 243.061	96 **Cm** Curium 247.070	97 **Bk** Berkelium 247.070	98 **Cf** Californium 251.080	99 **Es** Einsteinium [254]	100 **Fm** Fermium 257.095	101 **Md** Mendelevium 258.1	102 **No** Nobelium 259.101	103 **Lr** Lawrencium [262]

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